



ABSTRACT

What is 3DAgent?

An embodied, conversational AI agent built on **Matterport digital twins**. It fuses large language models, multimodal computer vision, and the Matterport SDK to turn passive 360° tours into spaces you can **talk to, scan, navigate, and maintain** — automating asset cataloguing and facility audits for non-technical users.

Problem & Objectives

Traditional 3D spaces are **passive and disconnected**; facilities teams spend hours cataloguing assets by hand from floor plans. 3DAgent maps natural-language intent to action, auto-catalogues assets from a screenshot, and closes the loop: **detect** → **tag** → **query** → **report** → **triage** → **resolve**.

14

Intent classes
in the router

93.3%

Intent classification
accuracy (30 cmds)

0.63

Asset detection
MAE (avg)

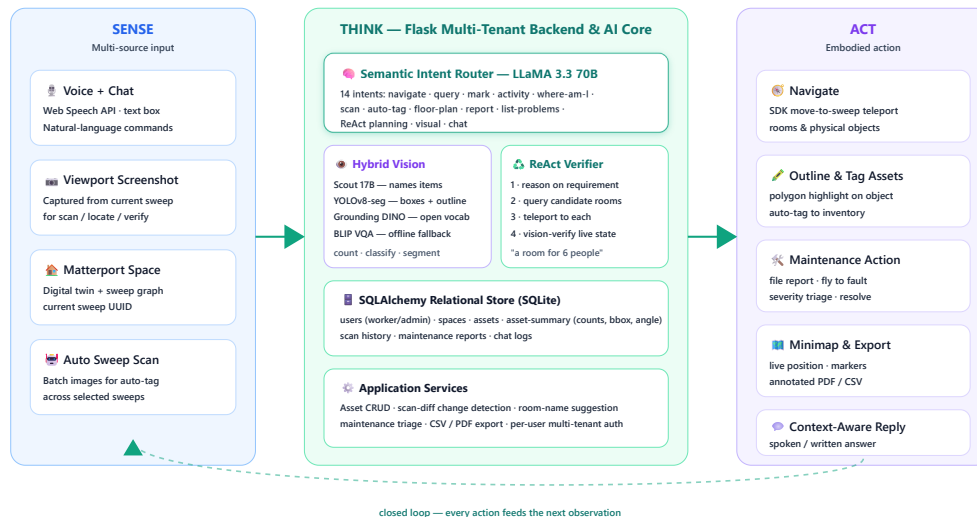
5

Vision / LLM
models orchestrated

SYSTEM ARCHITECTURE · SENSE → THINK → ACT

3DAgent — System Architecture

Embodied AI loop over a Matterport digital twin · SENSE → THINK → ACT



KEY FEATURES



NL & Voice Navigation

"I want to cook" or "take me to the nearest fire extinguisher" — Web Speech API enables hands-free control.



AI Vision Asset Detection

Scout + YOLOv8-seg + Grounding DINO detect, count, classify, **outline**, and auto-tag assets from a viewport.



ReAct Agentic Verification

Reasons on a request, finds candidate rooms, teleports, and **verifies live state** by vision before recommending.



Maintenance & Triage

Voice-file a fault pinned to a sweep; severity ranking, status lifecycle, and fly-to faulty equipment.



Smart Asset Inventory

Full CRUD, per-area counts, **scan-history change detection**, one-click CSV / PDF export.



Floor-Plan Minimap

Live position tracking with asset markers; export annotated floor plans as PNG / PDF.

NOVELTY & INVENTIVENESS

- 1. **Digital Twin × LLM fusion** — a 70B LLM drives a real Matterport camera, so it *teleports* on reasoning, not menu clicks; one chat box becomes a 14-class semantic router.
- 2. **Hybrid open-vocabulary vision** — Llama 4 Scout *names* any item (open vocab), then YOLOv8-seg + Grounding DINO ground each name with boxes and **edge-fitting polygon outlines** — more reliable than asking an LLM for coordinates.
- 3. **ReAct agentic verification** — sense → think → act → **verify**: the agent visits each candidate room and confirms its current state with live vision before answering.
- 4. **Voice-filed, location-pinned maintenance** — "report chair #1 has a broken leg" extracts asset + severity and pins the issue to an exact sweep for later fly-to repair.
- 5. **Resilient by design** — cloud Groq inference for speed, with a local BLIP VQA + YOLO path that keeps detection working with no cloud key.

USEFULNESS & APPLICATION

 **Who & what it replaces**

Built for facilities & operations managers, factory/warehouse supervisors, lab & hospital asset coordinators, and real-estate / retail teams. Replaces **clipboard audits, static PDF floor plans, spreadsheet registers, and disconnected maintenance tickets** — hours of cataloguing collapse into a few screenshots.

 **Where it applies**

Conversational Q&A and instant inventory across many settings:

Industrial / Warehouse

Corporate Office

Education

Healthcare & Labs

Real Estate

Retail

TECHNOLOGY STACK & HARDWARE

 **Tech Stack**

Language AI	LLaMA 3.3 70B (intent + chat) via Groq
Vision (cloud)	Llama 4 Scout 17B — open-vocab naming & room labelling
Vision (local)	YOLOv8 / YOLOv8-seg, Grounding DINO, BLIP VQA (fallback)
ML runtime	PyTorch · Ultralytics · HF Transformers
Backend	Flask + SQLAlchemy · SQLite · FPDF2
Frontend	HTML5/CSS3 · JS ES6+ · Jinja2 · Web Speech & Canvas API
3D platform	Matterport SDK (twin + sweep navigation)

 **Hardware Specification**

- **Capture:** Matterport Pro2 / Pro3 (or compatible 360°/LiDAR camera) to build the digital twin.
- **Server / dev:** i7 / Ryzen 7 · 16 GB RAM · NVIDIA CUDA GPU ≥ 6 GB VRAM for fast local CV (*CPU-only also works*) · ~5 GB for model weights · Python 3.10+.
- **Cloud:** Groq API serves LLaMA 3.3 70B + Scout 17B — no local GPU needed for the LLM/VLM path.
- **Client:** any modern browser (Chrome/Edge) with WebGL + microphone.

RESULTS · TEAM · SDG

 **Results & Evaluation**

93.3%

Intent accuracy
(30 commands)

0.63

Asset detection
MAE (avg)

7

Automated
evaluators

Unit evaluators (label match, vision parser, activity map, scan-diff, latency) pass 100%; intent & ReAct parser run as live-API integration tests. DB endpoints respond in ~1–2 ms (p95).

 **Project Team**

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 **Sustainable Development Goals (SDG)**

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Industry, Innovation & Infrastructure

— brings AI and digital-twin infrastructure to building and facility operations.

11

Sustainable Cities & Communities

— smarter, safer, more accessible building management with rapid hazard and fault reporting.

12

Responsible Consumption & Production

— accurate asset registers and condition tracking extend equipment life and cut waste.

8

Decent Work & Economic Growth

— location-pinned maintenance improves worker safety and reduces downtime.